Program: Artificial Intelligence & Machine Learning	B. Tech.	Semester: VI
Course: Deep Learning (DJS23ACPC601)		
Course: Deep Learning Laboratory (DJS23ALPC601)		

Pre-requisite: Natural Language Processing, Machine Learning.

Course Objectives:

1. To introduce fundamental concepts of artificial neural network and different learning algorithms: supervised and unsupervised neural networks
2. Develop in-depth understanding of the key techniques in designing Deep Network and GAN.
3. To expose Deep Network based methods to solve real world complex problems.
4. To explore applications and challenges in deep learning

Course Outcomes: Students will be able to

1. Understand the fundamentals of deep neural networks and their training mechanisms.
2. Apply optimization and regularization techniques to improve model performance.
3. Design and implement CNN models for supervised learning tasks.
4. Develop solutions for sequence learning applications using recurrent networks.
5. Analyze unsupervised learning techniques for dimensionality reduction and data reconstruction.
6. Evaluate recent trends in adversarial networks and generative models.

Deep Learning (DJS23ACPC601)		
Unit	**Description**	**Duration**
1	**Supervised Learning Networks** **Feedforward DNN** Perceptron: Representational power of Perceptron, The Perceptron Training Rule, Multilayer perceptron: Delta training rule; Multilayer Networks: A differentiable Threshold Unit (Sigmoid Neurons), Representational Power of Feedforward Networks; Activation functions: Tanh, Logistic, Linear, Softmax, ReLU, Leaky ReLU, Loss functions: Squared Error loss, Cross Entropy, Choosing output function and loss function	7
2	**Optimization** Learning with backpropagation: EBPTA, Learning Parameters: Gradient Descent (GD), Stochastic and Mini Batch GD, Momentum Based GD, Nesterov Accelerated GD, AdaGrad, Adam, RMSProp, Convergence and local minima, stopping criteria. **Regularization:** Regularization for Deep Learning: Parameter Norm Penalties, Dataset Augmentation, Noise Robustness, Early Stooping, Sparse Representation, Dropout.	6

3	**Convolutional Neural Networks:** Convolution operation, Padding, Stride, Relation between input, output and filter size, CNN architecture: Convolution layer, Pooling Layer, Weight Sharing in CNN, Fully Connected NN vs CNN, Variants of basic Convolution function, 2D Convolution **ConvNet Architectures:** LeNet: Architecture, AlexNET: Architecture, ResNet : Architecture, ConvNeXt, EfficientNET, **Applications:** object detection and recognition tasks, medical image analysis, image classification	6
4	**Sequence Modelling:** Sequence Learning Problem, Unfolding Computational graphs, Recurrent Neural Network, Bidirectional RNN, Backpropagation Through Time (BTT), Limitation of "vanilla RNN", Vanishing and Exploding Gradients, The Long Short-Term Memory, GRU, Deep recurrent Networks. **Applications:** Sentiment analysis, stock prices or market trends	7
5	**Unsupervised Learning Networks:** Kohonen Self-Organizing Feature Maps – architecture, training algorithm **Autoencoders:** Introduction, comparison with PCA, Linear Autoencoder, Undercomplete Autoencoder, Overcomplete Autoencoders, Regularization in Autoencoders, Denoising Autoencoders, Sparse Autoencoders, Contractive Autoencoders, Variational Autoencoders (VAEs) **Applications:** image compression, feature extraction, risk assessment and fraud detection	8
6	**Adversarial Networks** Generative Vs Discriminative Modeling, Generative Adversarial Networks (GAN) Architecture, GAN challenges: Oscillation Loss, Mode Collapse, Uninformative Loss, Hyperparameters, Tackling GAN challenges, Wasserstein GAN, Cycle GAN, Neural Style Transfer **Diffusion Models:** Introduction, Comparison with GANs **Applications**: Image synthesis or style transfer, Data Augmentation	8
	TOTAL	42

Books Recommended:

Text Books:

1. Dive into Deep Learning: Asaton Zhang, Zhacary Lipton, Mu Li and Alex Smola, December 2023
2. Understanding Deep Learning, Simon Prince, MIT Press, Dec2023
3. Simon Haykin, "Neural Networks and Learning Machines", Pearson Prentice Hall, 3rd Edition, 2010.
4. S. N. Sivanandam and S. N. Deepa, "Introduction to Soft Computing", Wiley India Publications, 3rd Edition, 2018.
5. M. J. Kochenderfer, Tim A. Wheeler. ―Algorithms for Optimization‖, IT Press.
6. David Foster, "Generative Deep Learning", O'Reilly Media, 2019.

7. Denis Rothman, "Hands-On Explainable AI (XAI) with python", Packt, 2020.

Reference Books:

1. Ian Goodfellow and Yoshua Bengio and Aaron Courville, "Deep Learning", An MIT Press, 2016
2. François Chollet, "Deep Learning with Python", Manning Publication, 2017.
3. Josh Patterson, Adam Gibson, "Deep Learning: A Practitioner's Approach", O'Reilly Publication, 2017.
4. Andrew W. Trask, Grokking, "Deep Learning", Manning Publication, 2019.
5. John D. Kelleher, "Deep Learning", MIT Press Essential Knowledge series, 2019.
6. Douwe Osinga. ―Deep Learning Cookbook‖, O'REILLY, SPD Publishers, Delhi

Web Resources Blogs and Websites:

1. Deep Learning: https://vlab.spit.ac.in/ai/#/experiments
2. Deep learning book https://www.deeplearningbook.org/
3. Deep learning all videos: https://www.cse.iitm.ac.in/~miteshk/CS6910.html
4. Deep Learning Specialization: https://www.coursera.org/specializations/deep-learning

Online Resources

1. Deep Learning, IIT Ropar NPTEL course by Prof. Sudarshan Iyengar,Dr. Padmavati https://nptel.ac.in/courses/106106184

Suggested List of Experiments:

Deep Learning Laboratory(DJS23ALPC601)	
Sr. No.	**Title of the Experiment**
1	Implement Boolean gates using perceptron.
2	Implement representation power of perceptron.
3	Implement backpropagation algorithm from scratch.
4	Train CNN Models for Image Classification Tasks.
5	Evaluate the Effect of Optimizer SGD on Model Performance.
6	Evaluate the Effect of Optimizer Adam on Model Performance.
7	Compare the Performance of PCA and Autoencoders on Dimensionality Reduction
8	Sequence Classification Using RNN or GRU (e.g., Sentiment Analysis or Activity Recognition).
9	Anomaly detection using Self-Organizing Network.
10	Compare the performance of PCA and Autoencoders on a given dataset.
11	Train Variational Autoencoders (VAEs) for Image Reconstruction.
12	Build Generative adversarial model for fake (news/image/audio/video) prediction.
13	Generate Synthetic Data Using Diffusion Models and Evaluate Results.
14	Mini Project

Minimum ten experiments from the above suggested list or any other experiment based on syllabus will be included, which would help the learner to apply the concept learnt.

Prepared by Checked by HoD Vice-Principal Principal